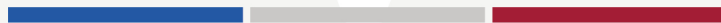




MIT SLOAN SCHOOL OF MANAGEMENT
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MODULE 2 UNIT 3
Media Set Video 1 Part 2 Transcript

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Engineering consistency for an inconsistent medium

PAUL MCDONAGH-SMITH: So, Dmitri, one of the puzzles for people who are new to agentic systems is often that LLMs are famously inconsistent. You know, the idea that you can ask the same question twice and you'll get two different answers. And yet our enterprises run, or hopefully run on consistency. How do you engineer consistency into an agentic workflow? What kind of checks and balances have you found genuinely valuable in production? And conversely, what is the most expensive check you've discovered, in retrospect, that you should have built in from the beginning?

DMITRI TCHEREVIK: Yeah, it's a very good question. And you point it there that if there's the same problem, we're dealing with people. People are famously inconsistent. You ask the same person the same question, you get the different answers, right?

And so, the same approach that you use to achieve consistency with people can be used to achieve consistency with AI. So this task decomposition, you make the task simpler. Then education.

But models, they are very good at following examples. So if you give an example, "This is what I would like you to produce," AI model will do its best to follow that example and produce something that's consistent with your example. So it's, there are these terms – few-shot, multi-shot learning, right? So you can give it multiple shots. But then also, there are other tricks you can do. You can ask the same question multiple times. It's called self-consistency. And then you can take the average of the answers. So there are multiple approaches that an agentic workflow allows you to do.

So you can have one model generate an answer, and then you can have another model check that answer. So to answer your question about the most important check is check your references. So the model may produce an answer, may have made some references to the materials it used to deliver that answer. You can always have another model go and check the references, and then you reduce hallucinations. So I would say prompt engineering practices, examples, and then validation. Build that into your agentic workflows.

MCDONAGH-SMITH: That final point around, have you found yourself working with a client where you've thought, "Okay, there was one point that we should have actually built in from the beginning?" Is there a lesson learned? Anything that you would share with our participants here to help them kind of avoid some of the potholes that might be kind of present in this type of activity?

TCHEREVIK: Yeah, would definitely check the references. Have a step there that is validating the references because you can ask a model to deliver references to everything that it generates. And then you can have another model go in and cross-reference, make sure that those are valid. But then also, task decomposition in itself is a very valuable mechanism.

Let's say if you want to process 60 leads, right, you can give them as a set to a model and say, "Go and process all 60." And then the model, I guarantee you, will get confused. But you can give them in groups of three. You can give them just one lead at a time, and that will improve the precision and consistency.

Encoding a methodology end-to-end

MCDONAH-SMITH: Dmitri, participants will leave this module having to translate their own working methodologies into agentic workflows. Could you perhaps walk us through one engagement where you helped a non-technical team do exactly that? From how you elicited what they actually do, through to where the encoded version ended up differing from how they had been doing it manually, whatever workflow, whatever process. What did the methodology have to give up in order to be encodable? And what did it gain?

TCHEREVIK: Yeah. So, one good example is discovery phase in a management consulting engagement. So, this was a management consulting company. They are all non-technical people. They're doing HR consulting. So very often, always in the beginning of their engagement, they go and they interview stakeholders. So, the stakeholders may be Chief Executive, Chief Financial Officer, then also line managers, HR managers. So they do like dozens of those interviews.

And then there is an analysis phase. So, they take all the interviews, and they analyze them cross-sectionally. And the whole process takes months, as you might imagine.

So first, AI is very good at doing cross-sectional analysis. So that was the first agent that they built. Non-technical team, they prompted the agent, they learned how you can give that agent dozens of interviews, and then ask for consistencies, inconsistencies, tensions, opportunities.

And that was just a a-ha moment because, within minutes, it was able to produce something that it would take weeks for them to do previously, manually.

And the next step, so this was just simplifying the existing process. And the next step, now that you have AI, can we do more? Can we interview 60 people instead of 16? And so with AI, you can build chatbots. Chatbots that can walk a person through an interview. And now they can certainly interview not only line managers, but maybe also individual contributors and greatly expand the scope of their discovery and then use the same cross-sectional analysis to now crystallize knowledge across a much broader audience or sample.

So it's something that they have not been able to do before. They're all non-technical people. Of course, they lost some personal touch, like chatbot is doing some of the interviews. So they lost that, but they gained the breadth.

MCDONAGH-SMITH: Are there methods that you employ to help people get over this little bump?

TCHEREVIK: Yeah, so my first advice is always just start using it. Because very often people make conclusions about AI without spending enough time with So, and that addresses two issues. So one of them is that they understand how helpful it can be in all kinds of tasks, but then they also start bumping into limitations, and that gives them

confidence. Because now they feel less threatened by AI. They know that it cannot do certain things that I do very well.

And now you get into this human–AI collaboration space. I think that competency, that AI competency and comfort, must be built throughout the organization. Unlike some of the other process reengineering efforts that are mostly driven top-down. Here, you must have top-down motion, but also bottom-up motion.

MCDONAGH-SMITH: How should our leaders be actually putting themselves forward, actually transmitting their confidence in the technology?

TCHEREVIK: Well, first of all, they should be using AI themselves. Many people don't, but as a leader, if you want to achieve AI transformation, you must be using AI and letting people know that you're using AI. So that's one thing. Another thing is that it's communicating that it's okay to use AI to accomplish your task. Because there are some people who are very comfortable using AI. And believe it or not, there are some people who use it in secret because they are not sure that if they generated output using AI, can they say that they used AI. So that needs to be, I think, somehow communicated that AI is a powerful tool. Everybody should be using it.

MCDONAGH-SMITH: Yeah, that's the point, the connection, the needle I was trying to thread there between almost the role-modeling of leadership and the permission-giving to others in the organization.

TCHEREVIK: That's right, yes. So then there must be policies. For some reason, for some tasks, AI may not be appropriate. Like, you cannot be making irreversible decisions based on AI recommendation because of certain limitations. But for some tasks, AI is a great tool, and it should be used. So I guess some kind of guardrails, governance policies, and that's the role of senior leaders, actually, to define those policies for using AI.

SPEAKER: Did you understand all the concepts covered in this video? If you'd like to review any of the content, please click on the chapter menu.